

# Displays

D1:

$$\hat{R}(\theta) = \{x \in \mathbb{R}^{n_x} : g(x, \theta) \leq 0\} \quad (1)$$

D2:

$$e \equiv V(\hat{R}(\theta)) \equiv \mathbf{P}_{\delta \sim \Delta} \{g(\delta, \theta) > 0\} \equiv \mathbf{P}_{\delta \sim \Delta} \{\delta \notin \hat{R}(\theta)\}. \quad (2)$$

D3:

$$\hat{e} \equiv \hat{V}(\hat{R}(\theta)) \equiv \mathbf{P}_{\delta_s \sim \Delta} \{\delta_s \notin \hat{R}(\theta)\} \quad (3)$$

D4:

$$\equiv \frac{1}{M} \sum_{i=1}^M \mathbb{1}_{\hat{R}(\theta)}(\delta_s^{(i)}) \quad (4)$$

D5:

$$\mathbb{1}_{\hat{R}(\theta)}(\delta_s^{(i)}) = \begin{cases} 1 & \text{if } \delta_s^{(i)} \notin \hat{R}(\theta) \\ 0 & \text{else.} \end{cases} \quad (5)$$

D6:

$$\begin{aligned} & \underset{\theta}{\text{minimize}} && \text{Vol}(\theta) \\ & \text{subject to} && g(\delta^{(i)}, \theta) \leq 0, i = 1, \dots, N \\ & && \theta \in \mathbb{R}^{n_\theta}. \end{aligned} \quad (6)$$

D7:

$$\overline{\text{Bin}}(\hat{k}, M, \beta) = \max_e \left\{ e : \text{Bin}(\hat{k}, M, e) \geq \beta \right\}, \text{ where} \quad (7)$$

D8:

$$\text{Bin}(\hat{k}, M, e) = \sum_{j=0}^{\hat{k}} \binom{M}{j} e^j (1-e)^{M-j}. \quad (8)$$

D9:

$$\mathbf{P}\{V(\hat{R}(\theta)) > \overline{\text{Bin}}(\hat{k}, M, \beta)\} \leq \beta. \quad (9)$$

D10:

$$\overline{\text{Bin}}(\hat{k}, M, \beta) \leq \frac{\hat{k}}{M} + O\left(\sqrt{\frac{\log(\frac{1}{\beta})}{M}}\right), \quad (10)$$

D11:

$$\mathbf{P}_{(\{\delta^{(i)}\}_{i=1}^N, \{\delta_s^{(i)}\}_{i=1}^M)} \{V(\hat{R}(\theta)) > \overline{\text{Bin}}(\hat{k}, M, \beta)\} \leq \beta.$$

D12:

$$\begin{aligned} & \underset{\mu, \sigma}{\text{minimize}} && \sum_{i=1}^m \sigma_i^2 \\ & \text{subject to} && \sum_{i=1}^m e^{-\frac{1}{2} \frac{(\delta^{(j)} - \mu_i)^2}{\sigma_i^2}} - \gamma \geq 0, \quad j = 1, \dots, N, \\ & && \sigma \in [0, \infty)^m \end{aligned} \quad (11)$$

D13:

$$g(x, \theta) = \sum_{i=1}^m e^{-\frac{1}{2} \frac{(x - \mu_i)^2}{\sigma_i^2}} - \gamma \quad (12)$$

D14:

$$\begin{aligned} \ddot{x} &= u_1 K \sin(\theta), \\ \ddot{h} &= -g + u_1 K \cos(\theta), \\ \ddot{\theta} &= -d_0 \theta - d_1 \dot{\theta} + n_0 u_2 \end{aligned} \quad (13)$$

D15:

$$\begin{aligned} x(0) &\in [-1.7, 1.7], h(0) \in [0.3, 2.0], \theta(0) \in [-\pi/12, \pi/12], \\ \dot{x}(0) &\in [-0.8, 0.8], \dot{h}(0) \in [-1.0, 1.0], \dot{\theta}(0) \in [-\pi/2, \pi/2], \end{aligned}$$

D16:

$$\begin{aligned} & \underset{\mu, \sigma}{\text{minimize}} && \sum_{\tau=0}^t \sum_{i=1}^m \sigma_i(\tau)^2 + \lambda \|\sigma_{avg} - \sigma_i(\tau)\|^2 \\ & \text{subject to} && \sum_{\tau=0}^t \sum_{i=1}^m e^{-\frac{1}{2} \frac{(\delta^{(j)}(\tau) - \mu_i(\tau))^2}{\sigma_i(\tau)^2}} - \gamma \geq 0, j = 1, \dots, N, \\ & && \sigma(\tau) \in [0, \infty)^m. \end{aligned}$$

D17:

$$A = \begin{bmatrix} -0.7 & -1.0 \\ 1.0 & -0.7 \end{bmatrix}. \quad (14)$$

D18:

$$\mathbf{P}_{x \sim \mu(B_2^d)} \{h(x) > 0\} \leq \varepsilon, \quad \mu(B_2^d) := \text{Unif}(B_2^d).$$

D19:

$$\{x \mid h_\delta(x) > 0\} = \{x \mid \delta/L > \|x\|_2\} =_{\text{a.e.}} \{x \mid x \in B_2^d(\delta/L)\},$$

D20:

$$\begin{aligned} \mathbf{P}_{x \sim \mu(B_2^d)} \{h_\delta(x) > 0\} &= \mathbf{P}_{x \sim \mu(B_2^d)} \{x \in B_2^d(\delta/L)\} \\ &= \frac{\text{Vol}(B_2^d(\delta/L))}{\text{Vol}(B_2^d(1))} = (\delta/L)^d. \end{aligned}$$

## Inline

I1:  $R = \{\Phi(t_1; t_0, x_0, d) : x_0 \in X_0, d \in D\}$

I2:  $X_0 \subseteq \mathbb{R}^{n_x}$

I3:  $D$

I4:  $d : [t_0, t_1] \rightarrow \mathbb{R}^{n_d}$

I5:  $\Phi : X_0 \times D \rightarrow \mathbb{R}^{n_x}$

I6:  $t_1$

I7:  $X_0$

I8:  $t_0$

I9:  $\hat{R}$

I10:  $\mu_{X_0}$

I11:  $\mu_D$

I12:  $\Delta$

I13:  $\delta^{(i)} = \Phi(t_1; t_0, x_{0i}, d_i), i = 1, \dots, N$

I14:  $x_{01}, \dots, x_{0N} \stackrel{i.i.d}{\sim} \mu_{X_0}$

I15:  $d_1, \dots, d_N \stackrel{i.i.d}{\sim} \mu_D$

I16:  $g : \mathbb{R}^{n_x} \times \mathbb{R}^{n_\theta} \rightarrow \mathbb{R}$

I17:  $\theta$

I18:  $\delta^{(1)}, \dots, \delta^{(N)}$

I19:  $\beta$

I20:  $\mathbf{P}\{V(\hat{R}(\theta)) > \epsilon\} \leq \beta$

I21:  $V(\hat{R}(\theta))$

I22:  $e$

I23:  $\hat{R}(\theta)$

I24:  $V(\hat{R}(\theta)) \leq \epsilon$

I25:  $\epsilon$

I26:  $M$

I27:  $\hat{e}$

I28:  $\text{Vol} : \mathbb{R}^{n_\theta} \rightarrow \mathbb{R}_{\geq 0}$

I29:  $R(\theta)$

I30:  $k$

I31:  $\hat{k}$

I32:  $\beta \in (0, 1]$

I33:  $\delta_s^{(i)} = \Phi(t_1; t_0, x_{0i}^s, d_i^s), i = 1, \dots, M$   
 I34:  $x_{01}^s, \dots, x_{0M}^s \stackrel{i.i.d}{\sim} \mu_{X_0}$   
 I35:  $d_1^s, \dots, d_M^s \stackrel{i.i.d}{\sim} \mu_D$   
 I36:  $\delta_s^{(1)}, \dots, \delta_s^{(M)}$   
 I37:  $\beta \in (0, 1)$   
 I38: minimize  $\text{Vol}(\theta)$   
 I39:  $\theta \in \bigcap_{i=1, \dots, N} \{\theta_i : g(\delta^{(i)}, \theta_i) \leq 0\}$   
 I40:  $\mathbf{P}\{V(\hat{R}(\theta)) > \epsilon\} \leq \beta$   
 I41:  $\epsilon := \overline{\text{Bin}}(\hat{k}, M, \beta)$   
 I42:  $e \mapsto \text{Bin}(k, M, e)$   
 I43:  $[0, 1]$   
 I44:  $\overline{\text{Bin}}(\hat{k}, M, \beta)$   
 I45:  $\hat{k} = 0$   
 I46:  $\overline{\text{Bin}}(0, M, \beta) \leq \frac{\log(1/\beta)}{M}$   
 I47:  $1/M$   
 I48:  $\log(\frac{1}{\beta})$   
 I49:  $\hat{k} > 0$   
 I50:  $\{\delta_s^{(i)}\}_{i=1}^M$   
 I51:  $N$   
 I52:  $\{\delta^{(i)}\}_{i=1}^N$   
 I53:  $N + M$   
 I54:  $\text{Vol}(\hat{R}(\theta)) = \sqrt{\sum_{i=1}^m \sigma_i^2}$   
 I55:  $\mu_i$   
 I56:  $\sigma_i$   
 I57:  $\gamma = 0.25$   
 I58:  $N + M = 3000$   
 I59:  $\beta = 10^{-9}$   
 I60:  $N = 3000$   
 I61:  $\ddot{x} = -\alpha y + x - x^3 + \gamma \cos(\omega t)$   
 I62:  $x, y \in \mathbb{R}$   
 I63:  $\alpha, \gamma, \omega \in \mathbb{R}$   
 I64:  $\alpha, \gamma$   
 I65:  $\omega$   
 I66:  $\alpha = 0.05, \gamma = 0.4, \omega = 1.3$   
 I67:  $x(0) \in [0.95, 1.05]$   
 I68:  $y(0) \in [-0.05, 0.05]$   
 I69:  $[t_0, t_1] = [0, 100]$   
 I70:  $\text{Vol}(\hat{R}(\theta))$   
 I71:  $m = 2$   
 I72:  $N = 10$   
 I73:  $N = 2990$   
 I74:  $N = 1500$   
 I75:  $M = 1500$   
 I76:  $\epsilon = 0.018$   
 I77:  $\epsilon = 0.035$   
 I78:  $\text{Vol}(\hat{R}(\theta)) = 1.55$   
 I79:  $x$   
 I80:  $h$   
 I81:  $x, h, \theta$   
 I82:  $u_1$   
 I83:  $u_2$   
 I84:  $g = 9.81, K = 0.89/1.4, d_0 = 70, d_1 = 17, n_0 = 55$   
 I85:  $u_1(t) = u_1$   
 I86:  $u_2(t) = u_2$

I87:  $\forall t \in [t_0, t_1]$   
 I88:  $u_1 \in [-1.5 + g/K, 1.5 + g/K], u_2 \in [-\pi/4, \pi/4]$   
 I89:  $[t_0, t_1] = [0, 5]$   
 I90:  $m = 3$   
 I91:  $\epsilon = 0.051$   
 I92:  $\text{Vol}(\hat{R}(\theta)) = 27.80$   
 I93:  $N + M = 2000$   
 I94:  $N = 1000$   
 I95:  $M = 1000$   
 I96:  $\epsilon = 0.0263$   
 I97:  $\hat{\mathcal{R}}(t) = \{\Phi(t_i; t_0, x_0, d) : \forall t_i \in [t_0, t], x_0 \in X_0, d \in D\}$   
 I98:  $[t_0, t]$   
 I99:  $d : [t_0, t] \rightarrow \mathbb{R}^{n_d}$   
 I100:  $\Phi : \mathbb{R} \times X_0 \times D \rightarrow \mathbb{R}^{n_x}$   
 I101:  $\hat{\mathcal{R}}(t)$   
 I102:  $t_i$   
 I103:  $\mu$   
 I104:  $\sigma$   
 I105:  $\lambda$   
 I106:  $\gamma$   
 I107:  $\delta$   
 I108:  $\sigma_{avg}$   
 I109:  $\dot{x} = Ax$   
 I110:  $x_1(0), x_2(0) \in [1, 1.25]$   
 I111:  $[t_0, t] = [0, 10]$   
 I112:  $\epsilon = 0.144$   
 I113:  $\theta^*$   
 I114:  $\mathbf{P}\{g(\theta^*, x) \leq 0\} \geq 1 - \epsilon$   
 I115:  $\gamma \geq 0$   
 I116:  $g(\theta^*, x) \leq \gamma$   
 I117:  $h(\epsilon)$   
 I118:  $(\sup_x g(x, \theta) - g(x, \theta))$   
 I119:  $\mathbf{P}\{g(\theta^*, x) \leq h(\epsilon)\} = 1$   
 I120:  $g$   
 I121:  $p$   
 I122:  $h : B_2^d \mapsto \mathbb{R}$   
 I123:  $B_2^d(r) := \{x \in \mathbb{R}^d \mid \|x\|_2 \leq r\}$   
 I124:  $B_2^d := B_2^d(1)$   
 I125:  $L$   
 I126:  $\max_{x \in B_2^d} h(x) = L\epsilon^{1/d}$   
 I127:  $h_\delta(x) := \delta - L\|x\|_2$   
 I128:  $\delta > 0$   
 I129:  $\mathbf{P}_{x \sim \mu(B_2^d)}\{h_\delta(x) > 0\} = \epsilon$   
 I130:  $\text{Vol}(B_2^d(r)) = r^d \text{Vol}(B_2^d(1))$   
 I131:  $(\delta/L)^d = \epsilon$   
 I132:  $\delta = \delta_\star := L\epsilon^{1/d}$   
 I133:  $\max_{x \in B_2^d} h_{\delta_\star}(x) = h_{\delta_\star}(0) = L\epsilon^{1/d}$   
 I134:  $\blacksquare$   
 I135:  $\epsilon$   
 I136:  $(L/\gamma)^d$   
 I137:  $\max_{x \in B_2^d} h(x) \leq \gamma$   
 I138:  $1$   
 I139:  $\ell_\infty$   
 I140:  $|h(x) - h(y)| \leq L\|x - y\|_\infty$   
 I141:  $\ell_2$   
 I142:  $|h(x) - h(y)| \leq L\|x - y\|_2$

I143:  $\ell_p$